

Precedence of operations:

1. NOT
2. AND
3. OR

- Thus, in the expression

$$x_1 \cdot x_2 + \bar{x}_1 \cdot \bar{x}_2$$

First, we must generate the complements of x_1 and x_2 .

Then the product terms $x_1 \cdot x_2$ and $x_1' \cdot x_2'$ are formed.

Followed by (+) operation.

You can use parentheses to ensure that order as follows:

$$(x_1 \cdot x_2) + ((\bar{x}_1) \cdot (\bar{x}_2))$$

- Throughout this course, we will work with two fundamental types of tasks. Every question we study will belong to one of these two types:

1- Synthesis process: Given a description of the desired functional behavior, you then generate a circuit that realizes that behavior.

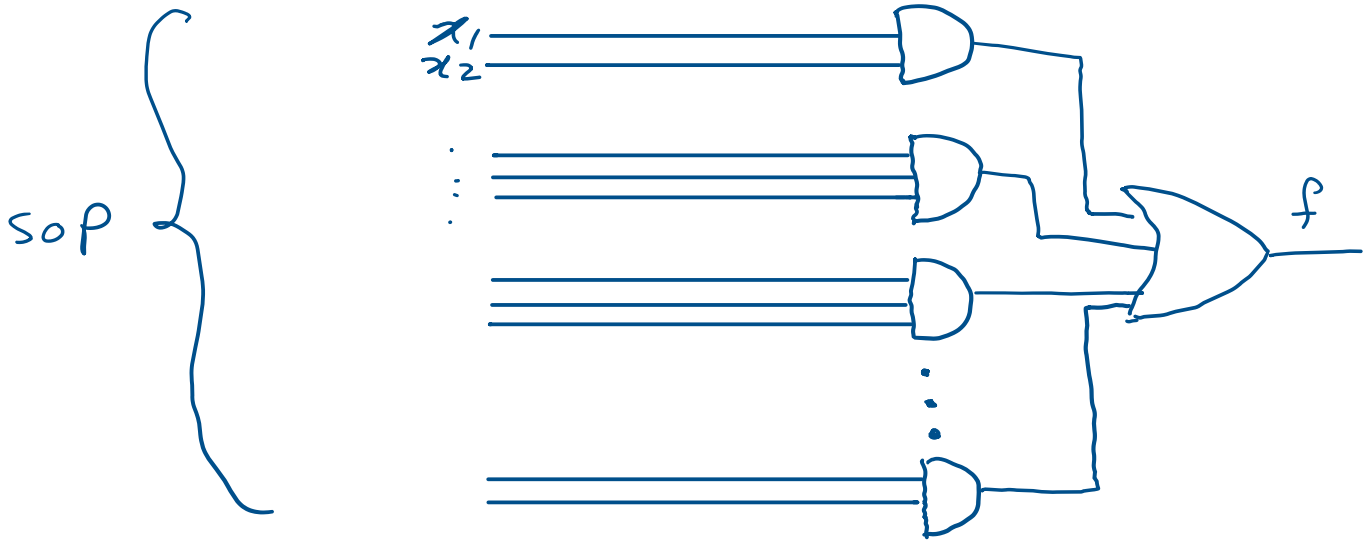
2- Analysis process: Given a circuit (a network of logic gates), you determine its functional behavior.

- Any two-level network of gates (circuit) can be represented in two forms:

1- Sum of Products (SOP)

2- Products of Sums (POS)

- Sum of Products (SOP): In general, it looks like the following circuit.



$$f(x_1, x_2, x_3) = x_1 \bar{x}_2 \bar{x}_3 + \bar{x}_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 + x_2 x_3 + \dots$$

↑
sop

- Draw a truth table for the following SOP expression:

$$f(x_1, x_2, x_3) = \bar{x}_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3 + \bar{x}_1 \bar{x}_2$$

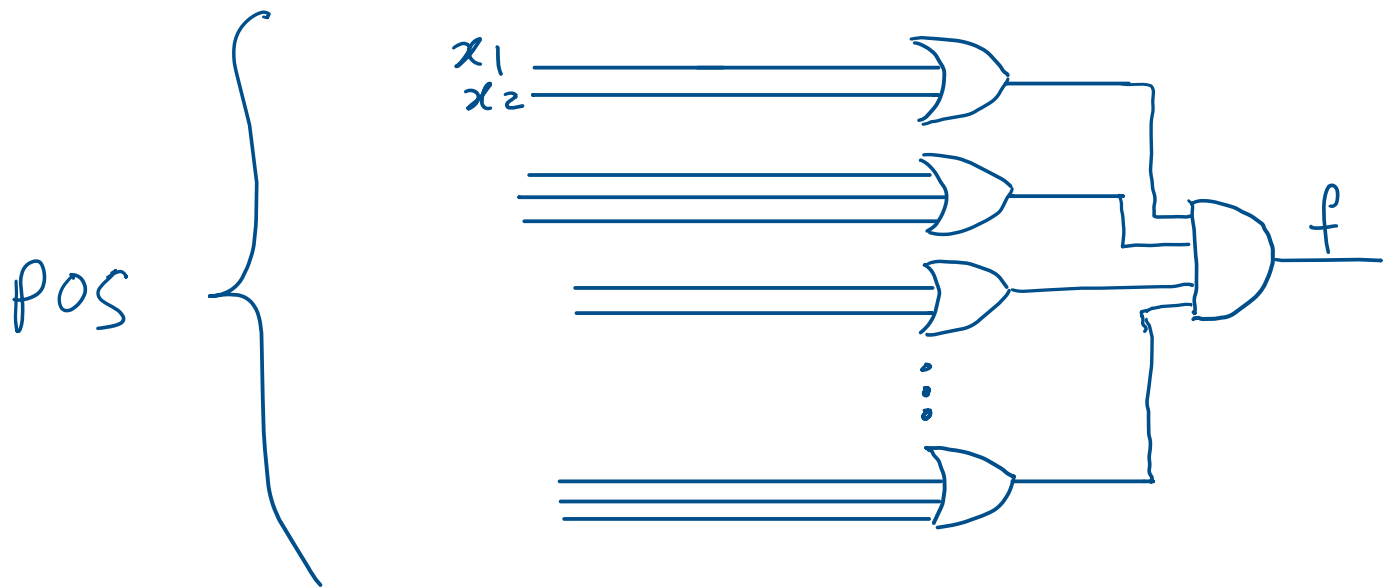
* Truth table

x_1	x_2	x_3	f
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- A **minterm** is an **AND** (or product) term that includes **every** variable exactly once, either in its true form (x) or complemented form (x').

- **Canonical SOP:** Is an SOP in which **every term** is a **minterm**.
- In other words, it is a sum of minterms.

- Product of Sums (POS): In general, it looks like the following circuit.



$$f(x_1, x_2, x_3) = (x_1 + \overline{x_2} + x_3) \cdot (\overline{x_1} + \overline{x_2} + x_3) \cdot (x_1 + x_2 + x_3) \cdot (\overline{x_2} + x_3) \cdot \dots$$

↑
pos

$$g(x_1, x_2, x_3) = (\bar{x}_1 + x_2 + x_3) - (\bar{x}_2 + \bar{x}_3)$$

x_1	x_2	x_3	g

A **Maxterm** is an **OR** (or sum) term that includes **every** variable exactly once, either in its true form (x) or complemented form (x').

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Canonical POS: Is an POS in which **every term** is a **Maxterm**.

- In other words, it is a product of Maxterms.

_Derive the canonical SOP for f' (i.e. $\overline{f}$)

x_1	x_2	x_3	f
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Example: Given the logical expression $f(a, b, c) = \sum m(1, 3, 5, 6)$,

- 1- Find the canonical POS form for f .
- 2- Find the canonical SOP form for f' (i.e. the complement of f)
- 3- Determine the simplest SOP form for f .